Week 14: Polygon triangulations and flips

Adult guide built from the problem files · Unpiloted. The tracing workflow, K-1 straightedge drawing and timing are untested.

This view repeats, by problem, the answers of the print guide, week-14-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py; its page references are to that guide, apart from those that name a student page (see Page numbers), and that guide also holds the pictured catalogs and route tables. The overview, preparation, launch, proofs and hints are there. Change both together; check_guide_copy.py compares them.

Problems and routes

Problem	K–1	2–3	4–5	route
Two fillings of each polygon	P1	P1		first
Every filling of the pentagon	P2	P2	P1	first
Three triangles or four in the hexagon	P3			first
Completions that keep a printed line	P4			return
Every result of one change	P5	P3	P2	K-1 return; 2-3 first; 4-5 first
A loop through the pentagon fillings	P6	P4		return
The flip map of the pentagon		P5	P3	2-3 return; 4-5 first
Fewest flips between two hexagon fans		P6		return
Every filling of the hexagon		P7		return
Three starts to one hexagon fan			P4	return
Fewest flips to two octagon fans			P5	return
A rule for the fewest flips to a fan			P6	return
Any two fillings are joined by flips			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P4–P6; Grades 2–3 P4–P7; Grades 4–5 P4–P7.

Page numbers. A key that names a student page counts the pages of the full band packet. The route packets keep the problem numbers but number their own pages; each route-packet page is this page of the full packet:

Route packet	its pages = full-packet pages
K–1 first	pp. 1–3 = pp. 1–3; p. 4 = p. 7
K–1 return	p. 1 = p. 1; pp. 2–5 = pp. 4–7
Grades 2–3 first	pp. 1–3 = pp. 1–3
Grades 2–3 return	p. 1 = p. 1; pp. 2–6 = pp. 4–8
Grades 4–5 first	pp. 1–3 = pp. 1–3
Grades 4–5 return	p. 1 = p. 1; pp. 2–5 = pp. 4–7

Key, one entry per problem

Two fillings of each polygon

K–1 P1 · Grades 2–3 P1 · first route · wording differs by band

Idea. Every filling of a convex polygon with n corners has $n-2$ triangles, however the lines are chosen.

K–1: Any two distinct legal fillings of each printed polygon suffice. For the pentagon, use $\{13, 14\}$ and $\{24, 25\}$. For the hexagon, use $\{13, 14, 15\}$ and $\{24, 25, 26\}$. Each pentagon has 3 triangles; each hexagon has 4.

Grades 2–3: Any two distinct legal fillings suffice. Hexagon examples: $\{AC, AD, AE\}$ and $\{BD, BE, BF\}$. Heptagon examples: $\{AC, AD, AE, AF\}$ and $\{BD, BE, BF, BG\}$. The counts cannot differ: every hexagon filling has 4 triangles and every heptagon filling has 5. Use page 3's splitting argument if children seek an explanation.

K–1: If a child repeats the same picture, compare the endpoints of every line. Turning the page around does not make a new filling: compare corner numbers, not looks.

Grades 2–3: Ask whether a thin or large triangle should still count as one. The task uses all triangles, regardless of size or shape.

Every filling of the pentagon

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P1 · first route · wording differs by band

Idea. A pentagon has exactly five fillings, the five fans: its two diagonals must share a corner, which then forces both.

K–1: There are five fillings, exactly the five fans on page 3 with letters replaced by numbers. Their diagonal sets are $\{13, 14\}$, $\{24, 25\}$, $\{13, 35\}$, $\{14, 24\}$, and $\{25, 35\}$. The page has six boards for five fillings, so one stays empty.

Grades 2–3: Exactly five pentagon fillings, displayed on page 3. The shared-corner argument there proves completeness.

Grades 4–5: The five pentagon fans on page 3 are the complete answer. The shared-corner proof or the root-edge classification proves no filling is missing.

K–1: Let the child keep each discovery on its own board; the packet's last page has spare boards. When ready to check completeness, ask whether the two lines meet at a corner and which corner it is.

Grades 2–3: Sort a child's collection by the corner shared by its two diagonals. Keep corner names fixed so that all five fans remain distinct.

Three triangles or four in the hexagon

K–1 P3 · first route

Idea. Any hexagon diagonal splits it into a triangle and a pentagon or two quadrilaterals, so every filling has $1 + 3 = 2 + 2 = 4$ triangles.

Three triangles are impossible; four are possible. For example, $\{13, 14, 15\}$ gives four. Any diagonal of a hexagon separates a triangle and a pentagon, or two quadrilaterals. Completing those sides gives $1 + 3 = 4$ or $2 + 2 = 4$ triangles. Thus changing the shape of the triangles cannot yield three.

Ask the child to point to any uncovered region in a three-triangle attempt. For a child seeking a reason that covers every attempt, cover one side of a chosen diagonal and count the two smaller pieces separately.

Completions that keep a printed line

K–1 P4 · return route

Idea. A kept diagonal splits the polygon into pieces filled independently, so the completions multiply: 2 for the pentagon, 2×2 for the hexagon.

Keeping 13 in the pentagon leaves a quadrilateral with two choices. Keeping 14 in the hexagon leaves two quadrilaterals, each with two independent choices: four completions. Only two pentagon completions exist, so one of the three pre-lined pentagons stays empty, and two of the six pre-lined hexagons stay empty; the spares keep the boards from giving the count away.

The required diagonal is printed on every board; if a child erases it, redraw it on a spare board from the packet's last page. If one hexagon side is always filled the same way, ask whether its quadrilateral has another filling. Combining each left choice with each right choice proves completeness.

Every result of one change

K–1 P5 · Grades 2–3 P3 · Grades 4–5 P2 · K-1 return; 2-3 first; 4-5 first route · wording differs by band

Idea. Erasing a diagonal leaves a convex quadrilateral with exactly one other diagonal, so a filling with $n-3$ diagonals has exactly $n-3$ one-change neighbours.

K–1: Each printed hexagon has exactly three one-change results, one for each of its three interior diagonals. The original filling itself is not a new result. Each small copy prints its start's lines in light gray: the child crosses out the gray line that goes and draws the new one, so every copy differs from its start by one line. Show this once at the table. Four copies for three results keep the count open. Handling the small copies. The copies are about 1.3 in across, and gray lines cannot be erased, so "erase" means cross out: the child slashes the gray line that goes and rules the new one, and a wobbly line that clearly joins the right two dots counts. If small hands cannot manage, the adult rules the line the child points to, the child works on a quarter tracing sheet over the large board, or the adult uses a whiteboard copy, where a line really can be erased. The child still chooses every change. Why the list is complete. Removing one interior diagonal merges exactly its two adjacent triangles into a convex quadrilateral. That quadrilateral has exactly one other diagonal. Everything outside it is unchanged. Thus each of the three original diagonals supplies exactly one neighbor, and there are no further legal one-line replacements.

Grades 2–3: First start $\{AC, AD, AE\}$ has neighbors $\{AC, AD, DF\}$, $\{AC, AE, CE\}$, and $\{AD, AE, BD\}$. Second start $\{AC, AE, CE\}$ has neighbors $\{AC, CE, CF\}$, $\{AE, BE, CE\}$, and $\{AC, AD, AE\}$. The six diagrams on page 5 show all results after replacing number labels by letters.

Grades 4–5: The two printed hexagon starts have the same six neighbors keyed in middle P3 and pictured in K–1 P5. Each start has exactly three possible flips. The two-triangle quadrilateral argument on page 5 proves completeness.

K–1: Trace the two triangles touching the line a child wants to erase. Ask for the boundary of their combined piece before asking which other line fills it. Do not erase a second line.

Grades 2–3: Choose one original diagonal; identify its two incident triangles. The other diagonal of their quadrilateral is the only legal replacement. Each small copy prints its start in light gray: cross out the gray line that goes and draw the new one.

Grades 4–5: A different-looking triangulation need not be one move away. Check that exactly one old diagonal disappeared and exactly one new one appeared. Each small copy prints its start in light gray: cross out the gray line that goes and draw the new one.

A loop through the pentagon fillings

K–1 P6 · Grades 2–3 P4 · return route · wording differs by band

Idea. Following the pentagon's five-cycle visits every filling once and returns in five flips, an odd number.

K–1: Yes. Name a pentagon filling by the corner where its two diagonals meet. The route $1, 3, 5, 2, 4, 1$ visits every other filling once and returns in five changes. The route in the opposite direction works too. Its complete diagonal sets are

$$\{13, 14\} \to \{13, 35\} \to \{25, 35\} \to \{24, 25\} \to \{14, 24\} \to \{13, 14\}.$$

Each arrow replaces exactly one line. The pentagon map on page 6 shows why all five fillings have been visited. The ring's top board is the fan at 1; four blank boards take four changes, and the last arrow home is the fifth. Only the first change is a choice; each later one is forced, since the other change goes back. Useful stopping point. Children who build and explain this five-step loop have encountered a genuine odd cycle. They do not need the word "parity" or the older packet's distance formula.

Grades 2–3: An odd return route is $F_A, F_C, F_E, F_B, F_D, F_A$, with five flips. The complete sets are the K–1 P6 route with letters replacing numbers. No shorter odd return is possible: the pentagon's flip map (Problem 5) has no loops or triangles. An immediate out-and-back takes two flips and does not meet the odd condition. Children find this route before drawing the map; Problem 5 then shows it as the map's cycle.

K–1: Lay the five Problem 2 boards beside the ring. Ask which of them can follow the current picture after one change. A child who reaches a filling already on the ring has gone back; look for the other change.

Grades 2–3: Ask children to count *moves*, not pictures, along their route. Six listed pictures represent five flips when the first is repeated at the end.

The flip map of the pentagon

Grades 2–3 P5 · Grades 4–5 P3 · 2-3 return; 4-5 first route · wording differs by band

Idea. The pentagon's five fillings and their flips form a five-cycle, so a filling can return to itself after an odd number of flips.

Grades 2–3: The complete pentagon map has exactly five joins: F_AF_C, F_CF_E, F_EF_B, F_BF_D, F_DF_A. There are no other joins. The revised student layout follows this cycle around the page. Each adjacent pair keeps one diagonal and replaces the other. Each of the five fillings has only two interior diagonals, hence exactly two possible flips; the displayed joins account for both and prove completeness.

Grades 4–5: The complete map is the five-cycle on page 6. The five-flip loop $F_A, F_C, F_E, F_B, F_D, F_A$ is an odd return. Its reverse is also valid. The graph is not bipartite: alternating two colors around the cycle makes the final edge join two same-colored vertices. Fidelity warning. A parity invariant from lozenge tiling flips does not transfer here. The objects and moves are different; a polygon five-cycle is a direct counterexample to an even-return claim.

Fewest flips between two hexagon fans

Grades 2–3 P6 · return route

Idea. Each flip adds at most one target diagonal, so a target with all three diagonals missing needs at least three flips, and three suffice.

The printed start is the B fan $\{BD, BE, BF\}$ and the target is the A fan $\{AC, AD, AE\}$. The minimum is three flips. Here is one shortest route, read left to right:

Step	Diagonals in the filling	Change just made
0	BD, BE, BF	Start
1	AE, BD, BE	BF $\to$ AE
2	AD, AE, BD	BE $\to$ AD
3	AC, AD, AE	BD $\to$ AC

Why three is best. None of the three target diagonals is present initially. One flip adds only one diagonal, so at least three flips are necessary. The displayed route achieves three. Counting missing target diagonals is a lower bound for arbitrary targets; it is always attainable for a fan target (page 11). A different simple route has four flips. The following five pictures are all different, so this meets the revised requirement to avoid visiting a filling twice.

Step	Diagonals in the filling	Change just made
0	BD, BE, BF	Start
1	BE, BF, CE	BD $\to$ CE
2	AE, BE, CE	BF $\to$ AE
3	AC, AE, CE	BE $\to$ AC
4	AC, AD, AE	CE $\to$ AD

Check the actual move. Two diagonal lists of length three are flip neighbors when they share exactly two diagonals. Merely rotating a page is not a move; nor may the child replace two diagonals at once.

For the shortest route, ask which target lines are missing and how many can appear in one move. For the second route, try a different first move. Undoing and redoing a move does not satisfy the no-repeated-fillings instruction. Routes are written on the ruled lines as lists of flips, such as "erase BF, draw AE" (Problem 5's example); the working hexagon is for trying moves.

Every filling of the hexagon

Grades 2–3 P7 · return route

Idea. Sorting by the triangle on one boundary edge splits the hexagon's fillings into $5+2+2+5=14$, with no omissions or repeats.

Student page 8 is labeled "Problem 7 (continued)". It supplies recording space for the same problem, not a second task. The complete hexagon collection has 14 fillings. Third corner B: five fillings. Diagonal BF cuts off triangle ABF, leaving a pentagon with five choices. Third corner C: two fillings. Triangle ACF leaves triangle ABC and quadrilateral CDEF. Third corner D: two fillings. Triangle ADF leaves quadrilateral ABCD and triangle DEF. Third corner E: five fillings. Triangle AEF leaves pentagon ABCDE. Why no omissions or repeats. A filling has exactly one triangle touching AF. Its third corner assigns it to exactly one group. The remaining polygon pieces can be filled independently; their complete smaller lists give $5+2+2+5=14$. No filling belongs to two groups.

Let children collect freely first. When a collection is hard to audit, ask which triangle touches AF in each picture. The student pages offer 17 recording outlines; the number of blank spaces is not the answer.

Three starts to one hexagon fan

Grades 4–5 P4 · return route

Idea. The number of target-fan diagonals missing from a start is a lower bound on the flips, and here each bound (3, 1, 2) is met.

The three starts, left to right, are $\{BD, BE, BF\}$, $\{AC, AE, CE\}$, and $\{AD, BD, DF\}$. The fourth filling, where the arrows point, is the A fan $\{AC, AD, AE\}$. Their minimum distances are 3, 1, and 2. Left start. Use the three-flip route on page 7: replace BF by AE, BE by AD, then BD by AC. It is shortest because all three target diagonals were missing. Middle start. Replace CE by AD. Exactly one target diagonal was missing. Right start. Replace BD by AC, then DF by AE. Exactly two target diagonals were missing.

Have a child mark which A-fan diagonals are already present, without erasing them. Ask how much that count can rise in one flip. A successful short route alone is not a proof that it is shortest. Each route is written as a list of flips on the ruled lines beside the blank hexagon, which is for trying moves.

Fewest flips to two octagon fans

Grades 4–5 P5 · return route

Idea. From the printed octagon filling the fans at A and E need 3 and 5 flips, the number of their diagonals missing at the start.

The printed start is $S=\{AC, AD, DF, DG, DH\}$. A fan at A has five diagonals $\{AC, AD, AE, AF, AG\}$; a fan at E has $\{AE, BE, CE, EG, EH\}$. The minimum distances are 3 and 5, respectively. Three-flip route to the A fan

Step	Diagonals in the filling	Change just made
0	AC, AD, DF, DG, DH	Start
1	AC, AD, AG, DF, DG	DH → AG
2	AC, AD, AF, AG, DF	DG → AF
3	AC, AD, AE, AF, AG	DF → AE

Five-flip route to the E fan

Step	Diagonals in the filling	Change just made
0	AC, AD, DF, DG, DH	Start
1	AC, AD, DG, DH, EG	DF → EG
2	AC, AD, DH, EG, EH	DG → EH
3	AC, AD, AE, EG, EH	DH → AE
4	AC, AE, CE, EG, EH	AD → CE
5	AE, BE, CE, EG, EH	AC → BE

Optimality certificates. The start already has two of A's five diagonals and none of E's five. Each flip can add at most one target diagonal. Therefore at least $5 - 2 = 3$ and $5 - 0 = 5$ flips are required. Both tables achieve their lower bounds, always adding a target diagonal and never deleting one. Children write each route on the ruled lines as a list of flips, such as "erase DH, draw AG", and try moves on the blank octagon.

Ask the child to choose a target corner and point to every diagonal already meeting it. If a proposed flip removes one of those, ask whether a different choice can increase the count instead. The general reason that such a choice exists is on page 11.

A rule for the fewest flips to a fan

Grades 4–5 P6 · return route

Idea. The fewest flips to the fan at A is $n - 3 - d_A$: each flip adds at most one A diagonal, and a flip that adds one always exists in a convex polygon.

For a convex n-gon, let d_A be the number of *interior diagonals* in the starting filling incident to A. The minimum number of flips to the A fan is

$$\boxed{n - 3 - d_A}.$$

Do not count the two boundary sides at A. For a triangle, $n - 3 = d_A = 0$, so the formula includes the zero-move case. Lower bound. Every filling has $n - 3$ diagonals. An A fan has all of them incident to A. One flip adds at most one new A diagonal, so at least $n - 3 - d_A$ flips are necessary. Why that bound can always be attained. Look at the union of all triangles touching A. If it is not the whole polygon, part of its boundary is an interior diagonal uw, with triangle Auw on one side and a triangle uwx on the other. Their union is a convex quadrilateral. Flip uw to Ax. The old edge is not incident to A, and the new one is. The union of triangles at A grows, and the number of A diagonals rises by exactly one. Repeat until every diagonal meets A. There are exactly $n-3-d_A$ increases. What constitutes an explanation. "I always add an A line" is a useful strategy but incomplete unless the child explains why such a flip exists whenever some A lines are missing. The boundary-of-the-shaded-region argument supplies that missing step. The lower bound and construction have different jobs, and both are needed for an exact minimum. Why convexity matters. In a nonconvex polygon, the union of two triangles need not be a convex quadrilateral; its other diagonal may leave the polygon. The guide's always-available flip argument is only for the strictly convex printed polygons.

First shade every triangle touching A on one concrete drawing. Ask where the shaded region could expand by changing one line. Once the child can do it twice, ask what would have to be true for the same move to be available again.

Any two fillings are joined by flips

Grades 4–5 P7 · return route

Idea. Any two fillings of a convex polygon are joined by flips: send both to the same fan and reverse one route.

The printed endpoints are $S = \{AC, AD, DF, DG, DH\}$ and $T = \{AE, BD, BE, EG, EH\}$. Here is a five-flip route.

Step	Diagonals in the filling	Change just made
0	AC, AD, DF, DG, DH	Start
1	AD, BD, DF, DG, DH	AC $\to$ BD
2	AD, BD, DG, DH, EG	DF $\to$ EG
3	AD, BD, DH, EG, EH	DG $\to$ EH
4	AD, AE, BD, EG, EH	DH $\to$ AE
5	AE, BD, BE, EG, EH	AD $\to$ BE

This route is also shortest, although the problem does not ask for that: the endpoints share no diagonal, so all five target diagonals must be introduced and one move adds at most one. Why any two fillings can be joined. Choose one fixed corner, say A. Page 11's growing-fan procedure turns the first filling into the A fan and the second filling into the same A fan. Reverse the second route and place it after the first. Every reversed flip is legal, since exchanging the two diagonals of a convex quadrilateral is reversible. This gives a route between the original fillings. It need not be shortest. An upper bound, not an exact distance. Passing through one fan uses $(n - 3 - d_A(S)) + (n - 3 - d_A(T))$

flips. For these endpoints A gives $3 + 4 = 7$, while the route above uses 5. This example shows why a connectivity construction should not automatically be called a shortest route.

If children can reach a fan but cannot connect arbitrary endpoints, ask whether a route can be followed backward. Be explicit that the same target corner must be used for both routes. A route can be recorded on the one working board as a list of changes, such as “erase AC, draw BD” from Problem 3’s example, with only the final filling drawn.